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ROOT CAUSE UNLOCKED · ULTIMATE GUIDE

Unlock Your Heart *Health*, Ultimate Guide

The Cholesterol, Blood Pressure, and Fatty Liver Connection Nobody Draws For You

A guide for adults who just got lab numbers or a blood pressure reading back that are trending the wrong way, who have not had a heart attack or stroke, and who want to understand what is actually driving those numbers instead of just being told to watch their diet.

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This edition is pending Dr. Seay's clinical sign-off. Every citation has been verified against its live PubMed record, including a check for retractions. The companion citation ledger lists each source and the claims that were investigated but left uncited.

BEFORE YOU START

A note before you start. This guide is educational information about the biology of cardiometabolic risk (how high cholesterol, high blood pressure, and fatty liver relate to each other) and the general categories of lifestyle and functional-medicine approaches used to address the underlying drivers. It is not individualized medical advice, and it does not replace a relationship with your physician or a clinical evaluation with me and my team. It does not diagnose anything. If you already have diagnosed heart disease, have had a heart attack, stroke, or stent or bypass procedure, or have diagnosed kidney disease, this guide is background reading only. Your cardiologist or nephrologist directs your care, and a direct conversation with our team is a better next step for you than working through this guide on your own. If you are currently taking a statin, a blood pressure medication, or a blood thinner, keep taking it exactly as prescribed. Nothing in this guide is a reason to stop, skip, or reduce any prescription medication. Talk to your prescribing physician before starting any new supplement, especially if you take any of the medications above, are pregnant or nursing, or have kidney disease.

Contents

Before Anything Else: When To Stop Reading And Call 911

This guide is written for someone with rising numbers, not someone in the middle of an emergency. Please read this section once, know it, and come back to the rest of the guide when it applies to you.

Call 911 or get to an emergency room immediately if you experience:

- Chest pain, pressure, squeezing, or tightness, especially if it lasts more than a few minutes or comes and goes
- Pain, pressure, or discomfort spreading to the jaw, neck, back, shoulder, or one or both arms
- Sudden shortness of breath, with or without chest discomfort
- Sudden weakness or numbness on one side of the face, arm, or leg, facial drooping, or sudden trouble speaking or understanding speech (the classic warning signs of stroke, sometimes taught as F.A.S.T.: Face drooping, Arm weakness, Speech difficulty, Time to call 911)
- A blood pressure reading of 180 systolic or higher, or 120 diastolic or higher (a hypertensive crisis range), particularly if paired with chest pain, shortness of breath, back pain, numbness, weakness, vision change, or difficulty speaking
- Sudden severe headache unlike any you have had before, sudden vision loss, or sudden loss of balance or coordination
- Nausea, cold sweat, or lightheadedness together with chest or upper body discomfort

None of these symptoms are something to sit with, look up, or wait out until your next appointment. They are something to act on immediately. This section exists because this guide will be read by people at genuine cardiovascular risk, and knowing these signs matters more than anything else in this document.

Everything below this point is for a different situation: understanding and addressing risk that is building quietly, before any of the above happens.

INTRODUCTION

The Numbers Nobody Connects For You

Maybe your cholesterol panel came back with a red flag next to LDL. Maybe your blood pressure has been creeping up at your last few checkups, and someone handed you a pamphlet about cutting salt. Maybe an ultrasound or CT scan done for something else mentioned “fatty liver” almost in passing, and nobody explained what that meant or why it was there.

Here is what usually does not happen in a fifteen minute visit: nobody sits down and shows you how those three things, high cholesterol, high blood pressure, and a fatty liver, are frequently the same underlying problem showing up in three different places. They get treated as three separate conversations, sometimes with three separate specialists, when in a large number of people they trace back to one shared root: how your body is handling insulin and blood sugar at the cellular level.

That is not a reason to distrust your cholesterol medication or your blood pressure prescription. Those medications treat real, measurable risk, and later in this guide we spend real time on exactly why that is true and why continuing them matters. It is a reason to understand what else is going on underneath the numbers, so that the terrain producing those numbers gets addressed too, alongside the medications, not instead of them.

This is the cluster of conditions that, taken together, is responsible for more deaths than almost anything else in modern medicine. That is precisely why it deserves a clear-eyed, non-alarmist explanation rather than either dismissal or panic. Below are nine keys. Each one explains a piece of the cardiometabolic picture, the mechanism behind it in plain language, and what you can do about it at an educational level. This guide does not diagnose you and it is not a treatment plan. It is the explanation that usually gets skipped.

KEY 1

Insulin Resistance Is the Shared Root of All Three

Three problems, one root. High cholesterol, high blood pressure, and a fatty liver are usually discussed as three unrelated problems that happen to show up in the same person as they age. In a large share of cases, they are not unrelated at all. They are three downstream expressions of the same upstream problem: cells that have stopped responding normally to insulin.

How one signal drives three outcomes. Insulin's job is to help your cells take up glucose from your blood. When cells become resistant to insulin's signal, most commonly driven by excess adiposity and a chronically high intake of refined carbohydrates, your pancreas compensates by producing more insulin to force the message through. Researchers describe insulin resistance as a root causative factor not only for metabolic syndrome itself but for the associated conditions that ride along with it: fatty liver disease, obesity-related type 2 diabetes, and atherosclerotic cardiovascular disease.[1] There is an active scientific debate about which comes first: whether insulin resistance itself drives the downstream damage, or whether the earlier problem is actually too much insulin (hyperinsulinemia), with insulin resistance developing afterward as a protective adaptation by tissues like the heart muscle trying to defend themselves against insulin overexposure.[1] Both camps agree on the practical point that matters here: disrupted insulin signaling is the mechanistic thread connecting dyslipidemia, elevated blood pressure, and fatty liver, not three separate coincidences.

Mechanistically, chronically elevated insulin pushes the liver to manufacture more triglyceride-rich particles (raising triglycerides and driving the small, dense LDL pattern discussed in Key 2), promotes sodium retention and sympathetic nervous system activation in the kidney and blood vessels (contributing to the blood pressure story in Key 6), and drives fat storage directly inside liver cells (the fatty liver story in Key 5). One disrupted signaling pathway, three visible symptoms.

How strong is that claim, really. This is the part worth pausing on, because "insulin is the root cause" is said constantly in wellness circles without anyone showing you the evidence, and it deserves better than a slogan.

The strongest available evidence here is not a diet study. It is a technique called Mendelian randomization, which uses the gene variants you were born with as a natural experiment. Because those variants are assigned at conception and cannot be influenced by your diet, your income, or your habits, they sidestep most of the confounding that makes ordinary nutrition research so difficult to interpret. It is about as close to a randomized trial as you can get for something nobody could ethically randomize.

Applied to insulin resistance, the answer is striking. Using 53 gene variants tied to the insulin-resistant phenotype, drawn from studies of up to 188,577 people for the exposure and 446,696 for the outcome, a one standard deviation increase in genetically predicted insulin resistance was associated with a **79 percent higher risk of coronary artery disease** (odds ratio 1.79), a 78 percent higher risk of myocardial infarction, and a 21 percent higher risk of ischemic stroke, with no evidence of the statistical pleiotropy that would undermine the finding. The authors concluded that insulin resistance is causally associated with coronary artery disease and myocardial infarction.[2] A separate 2025 analysis using different data and a different method arrived at almost exactly the same effect size for fasting insulin, an odds ratio of 1.79 for coronary artery disease, and went further by identifying the specific circulating proteins carrying that effect.[3]

So the claim that insulin sits upstream of cardiovascular disease is not a wellness talking point. It is supported by the same class of causal evidence used to establish the cholesterol relationship itself.

And here is where honesty requires stopping. A sex-specific analysis of 392,010 people in the UK Biobank found the association between genetically predicted insulin and both myocardial infarction and angina held **in men but not in women**.[4] That is one study and it does not overturn the others, but a guide that quotes the good numbers and hides that one is not worth your money. If you are a woman reading this, the insulin work is still worth doing for every other reason in this book, and the cardiovascular claim specifically is less settled for you than the headline suggests.

Insulin versus cholesterol is a false fight. This is the most useful thing in this chapter. The two positions get argued as though only one can be true, and they are actually the same story told at two different points in the same chain.

When researchers compared the lipid measurements head to head using that same genetic method, **apolipoprotein B was the one that survived**. Analyzed individually, LDL cholesterol, triglycerides and ApoB all looked harmful. Analyzed together, only ApoB retained a robust effect, the estimate for LDL cholesterol collapsed and even reversed direction, and the apparent protective effect of HDL attenuated essentially to nothing once ApoB was accounted for. The conclusion was that ApoB is the predominant trait explaining the relationship between blood lipids and coronary heart disease.[5] ApoB, as Key 2 explains, is a count of atherogenic particles rather than a measure of the cholesterol carried inside them.

Now put the two findings side by side. Insulin resistance drives the liver to manufacture more of exactly those particles. And in the proteomic analysis above, one of the mediators identified between insulin and coronary artery disease was the LDL receptor pathway itself.[3] **Insulin resistance is upstream. Particle number is the mechanism. Arterial damage is the outcome.** Anyone telling you it is one or the other is defending a position rather than describing the biology.

Which also makes the honest boundary of this chapter clear: insulin resistance is not the only road in. Familial hypercholesterolemia and elevated Lp(a) are genetically determined, they raise particle-driven risk in people who are lean and insulin sensitive, and no amount of metabolic work removes them. That is precisely why Key 2 asks you to measure them.

So why has nobody measured your fasting insulin. If this is genuinely upstream, the obvious question is why it is not on the standard panel. The reasons are structural rather than sinister, and knowing them helps you ask better questions.

First, **the test itself is not standardized between laboratories.** Insulin immunoassays show significant relative differences from one platform to another, and work on harmonizing them found that recalibration improved agreement at middle and high insulin levels **but not at low ones**, which is exactly the range where early detection would matter most.[6] A test without a universal number is a test that guidelines struggle to set a threshold for.

Second, **the diagnostic categories are built on glucose, not insulin.** Prediabetes and diabetes are defined by blood sugar and HbA1c. As described above, insulin climbs for years to hold glucose normal, so a system that screens on glucose is calibrated to find the problem only after the compensation has begun to fail.

Third, **fasting insulin is not in the cardiovascular risk calculators.** The equations your physician uses to make prevention decisions take cholesterol, blood pressure, smoking, age, and diabetes status. A marker that is not in the equation cannot change the estimate, and a marker that cannot change the estimate does not change the recommendation, however real it is.

Fourth, **the three downstream problems each have an owner and the upstream one does not.** Lipids, blood pressure, and the liver belong to different specialties, each with its own guideline and its own drug. Insulin resistance sits underneath all three and is nobody's assigned territory.

And fifth, worth saying plainly given how much of this book rests on published trials: **no company profits from proving that your insulin sensitivity improved.** Drug trials get funded because someone can sell the result. That asymmetry does not make the drug evidence wrong, and this book cites a great deal of it. It does mean the evidence base is far thicker on one side of this comparison than the other, and you should read that imbalance as a fact about how research gets funded rather than as a fact about your body.

None of this is an argument against treating cholesterol or blood pressure. It is an argument for asking one more question at your next appointment.

Where this points you next. This is why addressing insulin sensitivity, through nutrition pattern, movement, sleep, and in some cases targeted nutrient support, tends to move all three numbers at once rather than requiring three separate unconnected interventions. The rest of this guide walks through each expression of that shared root in more detail, and Key 9 covers the nutrition and movement patterns with real evidence behind improving insulin sensitivity itself.

KEY 2

What Your Standard Lipid Panel Does Not Tell You

What the same LDL number can hide. A standard cholesterol panel reports total cholesterol, LDL, HDL, and triglycerides. That panel has real value, and nothing in this guide is a reason to think otherwise. But two people can have the identical LDL number on paper and carry meaningfully different cardiovascular risk, because LDL cholesterol is a measurement of the cholesterol cargo inside a set of particles, not a direct count of the particles themselves.

What ApoB actually counts. Every atherogenic particle (the ones capable of lodging in your artery wall) carries exactly one molecule of a protein called apolipoprotein B (ApoB). That makes ApoB a direct count of the number of particles in circulation, rather than an estimate of the cholesterol they happen to be carrying that day. Because the amount of cholesterol packed into each particle varies from person to person, someone can have many small, cholesterol-poor particles (a high ApoB, higher risk) while their calculated LDL-C looks unremarkable, or the reverse.[7] This is exactly why professional cardiology and lipid societies increasingly reference ApoB alongside, not instead of, standard LDL-C: it resolves cases where the standard panel and the true particle burden disagree.

Lipoprotein(a), often written Lp(a), is a distinct, LDL-like particle that carries its own additional protein tag, making it more prone to promoting both plaque buildup and clotting. It is measured on request, not as part of a standard panel. Roughly seventy to ninety percent of the variation in Lp(a) levels between people is genetically determined, which means it is one of the few cardiovascular risk markers that is essentially set from an early age and does not move much with diet or lifestyle.[8] It also does not come down with statin therapy, since statins work on a different pathway. That does not mean it is unmanageable information, it means it changes how aggressively the other, modifiable risk factors (ApoB, LDL-C, blood pressure, inflammation) are worth managing in someone who has it.[8] Lp(a) is worth checking once, since it does not meaningfully change over your lifetime once you are past early childhood.

A third figure worth knowing is your triglyceride-to-HDL ratio, calculated from numbers already on a standard panel. It functions as an inexpensive stand-in for insulin resistance itself, the shared root from Key 1. Research reviewing decades of data across tens of thousands of people found average cutoffs of roughly 2.5 for women and 2.8 for men, above which insulin resistance becomes substantially more likely, though the exact cutoff varies somewhat by ethnicity and sex.[9]

And a fourth number, homocysteine, deserves the most honest handling of anything in this chapter. Homocysteine is an amino acid your body produces and clears as part of the methylation cycle, a set of reactions that runs on vitamin B12, folate, and B6. When one of those vitamins runs short, when kidney or thyroid function is off, or when a common gene variant (MTHFR) slows the recycling enzyme, homocysteine backs up in the blood. That is what makes it genuinely worth measuring: it is an inexpensive window into B-vitamin status, kidney function, and methylation capacity, systems worth knowing about in their own right.

What it is not, on the best current evidence, is a cardiac number to chase for its own sake, and the story of how we know that is the mirror image of the statin story in Key 3. In observational data pooling 30 studies, people with lower homocysteine did have modestly lower risk, about 11 percent less ischemic heart disease and 19 percent less stroke per 3 umol/L difference, and even the researchers reporting that association called homocysteine "at most a modest independent predictor" of those outcomes.[38] Then two harder tests arrived. A Mendelian randomization analysis, the same genetic technique this book leans on for insulin and ApoB, deliberately gathered 19 unpublished datasets totaling 48,175 coronary cases and found that lifelong moderately elevated homocysteine, produced by the MTHFR variant, had little or no effect on coronary heart disease: an odds ratio of 1.02, statistically indistinguishable from no effect. The previously published literature showed a significant 1.15, and the authors attributed the gap to publication bias, meaning small positive studies got published while null results stayed in file drawers.[39] And when randomized trials actually lowered homocysteine with folic acid in more than 82,000 people, stroke fell by about 10 percent and overall cardiovascular events by about 4 percent, with the benefit concentrated in people who started out folate-deficient, while coronary heart disease did not budge.[40]

Notice what happened there, because it is the same lesson as Key 3 running in the opposite direction. The published homocysteine literature was biased toward the position wellness culture favors, exactly as industry funding tilted the statin literature toward the position drug manufacturers favor. This book applies one standard to both: the strongest evidence wins, whoever it disappoints.

So here is the working conclusion, and it is the same one we use clinically at Root Health: treat an elevated homocysteine as a signal to investigate B12, folate, B6, thyroid, and kidney status, not as a target proven to change cardiac outcomes on its own. A result above roughly 10 umol/L is worth that workup conversation with your physician. Functional medicine practices, ours included, often read against a tighter target near 7, and standard laboratory ranges run looser still, so ask which frame your result is being read against. Two boundaries on all of this. The genetic analysis above speaks only to moderate, lifelong elevation;[39] homocystinuria, the rare inherited disorder where levels run several times normal, is a different and genuinely dangerous condition that belongs under medical management. And if the workup does find a B12 or folate deficiency, correcting it matters for your nerves, your blood, and your brain regardless of what it does or does not do for your arteries.

What these numbers change. None of this means your standard lipid panel was wrong, or that LDL cholesterol does not matter. It means there are more precise ways to measure the same underlying risk, and functional medicine uses ApoB, Lp(a), the triglyceride-to-HDL ratio, and homocysteine to sharpen the picture your standard panel already started, not to replace it or argue it was mistaken. If you already know your LDL is elevated, these additional numbers help clarify how urgently and how the modifiable pieces need to be addressed, working alongside whatever your prescriber has already recommended.

KEY 3

What the Statin Evidence Actually Shows, and What It Does Not

Two loud extremes, both wrong. Statin messaging tends to arrive from two opposite extremes. One says the entire cholesterol story is a myth and you should quietly stop your prescription in favor of supplements. The other treats "you should be on a statin" as settled for nearly everyone with an elevated number, full stop. Neither extreme reflects the actual evidence. What the data shows, how strong it is, and what it means for you depends heavily on one specific fact: whether you have already had a cardiac event, or you are looking at a number trending the wrong way for the first time. Collapsing those two situations into one blanket answer is exactly what makes this topic feel more settled, or more suspect, than it actually is.

Where genetics and drug trials agree. Start with what is genuinely not in dispute. LDL cholesterol's role in causing atherosclerotic cardiovascular disease does not rest on drug trials alone. Mendelian randomization studies, which use naturally occurring genetic variants that lower LDL from birth as a kind of lifelong natural experiment immune to industry funding or early trial stopping, consistently find the same result as the drug trials: lowering LDL, by whatever mechanism, lowers cardiovascular risk in proportion to how much it is lowered and for how long it stays lowered. A 2017 consensus statement from the European Atherosclerosis Society, reviewing genetic, epidemiologic, and clinical trial evidence together, concluded plainly that this combined evidence "unequivocally establishes that LDL causes ASCVD."[28] That is why "cholesterol has nothing to do with heart disease" is not a defensible reading of the literature. It is a separate question from how large a benefit a specific person, at a specific risk level, actually gets from taking a statin, which is where the real nuance lives.

The broadest randomized evidence for statin therapy comes from a meta-analysis pooling individual data from over 174,000 participants across 27 trials, which found major vascular events fell by about 22 percent in men and 16 percent in women per 1.0 mmol/L LDL reduction, with reduced all-cause mortality in both sexes and no signal of increased cancer or non-cardiovascular death in either sex.[10] That trial population is a mix that skews toward higher average risk than a reader who is simply noticing a rising number for the first time, including a substantial share of people who already have documented vascular disease, which is exactly why the absolute size of that 22 and 16 percent benefit needs to be read through the two categories below rather than as one number that applies equally to everyone: first the concrete absolute numbers for people who already have cardiovascular disease, then for people who do not.

If you have already had a heart attack, a stroke, or a stent or bypass procedure, the evidence is at its strongest and least contested. This is called secondary prevention: treating someone who already has documented disease, rather than trying to prevent a first event. In the trial that first established this, 4,444 patients with angina or a prior heart attack were randomized to simvastatin or placebo and followed for a median of 5.4 years. In the placebo group, 12 percent died; in the simvastatin group, 8 percent died, a 4 percentage point absolute difference (relative risk 0.70, a 30 percent reduction). Major coronary events occurred in 28 percent of the placebo group versus 19 percent of the simvastatin group, a 9 percentage point absolute difference (relative risk 0.66, a 34 percent reduction).[29] One more fact belongs in the same breath, because this guide checks funding in every direction: that trial was funded by Merck, the manufacturer of simvastatin, as nearly every landmark statin trial was manufacturer-funded. The reason the finding stands anyway is that the same direction of benefit shows up in the pooled analyses run by publicly funded academic groups[10] and in the genetic evidence described above, which no company sponsored.[28] A difference of several percentage points, measured over a few years, in people who already have the disease, is what strong evidence looks like in this field. If this is your situation, staying on your statin is not a close call.

If you have not had a cardiac event and are simply watching a number trend upward, the picture changes, not because the drug works differently, but because your starting risk is much lower, so the same relative benefit produces a much smaller absolute one. A 2012 meta-analysis from the Cholesterol Treatment Trialists' Collaboration, examining its pooled trial dataset specifically in people at low baseline vascular risk, found the proportional risk reduction per 1.0 mmol/L LDL reduction was at least as large in the lowest-risk participants as in higher-risk ones (a relative risk of 0.62 in the very lowest risk category, meaning roughly a 38 percent reduction), genuinely good news on the relative side. Translated into absolute terms, though, for people with a 5-year risk of a major vascular event under 10 percent, that reduction worked out to about 11 fewer events per 1,000 people treated over 5 years.[30] Eleven people out of a thousand, over five years, is a real benefit. It is also a small one, smaller than a 30 or 38 percent relative-risk figure tends to convey on its own. That is the actual, honest size of the number a low-risk reader is weighing, and well-informed people reasonably land in different places once they see it stated this way, which is exactly why this is an individual conversation with your prescriber rather than a foregone conclusion in either direction.

This same relative-versus-absolute gap is why one widely cited statin trial deserves closer scrutiny than it usually gets. That trial enrolled people with already-low LDL cholesterol but elevated hs-CRP (Key 4), and it is commonly summarized as a 47 percent reduction in first heart attack, stroke, or cardiovascular death, a figure repeated often enough that it functions as shorthand for "statins work."[11] The actual event rates behind that percentage were 0.77 per 100 person-years in the statin group versus 1.36 per 100 person-years in the placebo group, an absolute difference of about six-tenths of one percentage point per year.[31] That is a real effect, not a null one. But three things about this specific trial matter before it gets treated as a pillar of the case for statins. It was stopped early on a prespecified rule after a median of only 1.9 years of follow-up, and trials stopped early on their most favorable-looking interim result tend to overstate the true effect size. Independent reviewers have specifically pointed to the trial's strong commercial sponsorship as a plausible source of bias in how it was conducted and interpreted.[32] And the trial's lead investigator has disclosed being a co-inventor on patents related to the inflammatory biomarker testing used to select who could enroll in the first place.[33] None of that makes the finding false. It means this particular trial is the weakest possible anchor for the broader statin case, and a 47 percent headline without the underlying rate or these caveats is precisely the kind of framing that fuels distrust, and on this specific trial, that distrust has a real basis.

Two more things deserve honest treatment here, because patients ask about them before almost anything else, and a guide that leaves them out is not giving you the full picture. First, statins measurably raise the risk of new-onset type 2 diabetes. This is not a fringe claim. It is in the FDA-required product label, and it comes from the statin trialists' own pooled data. A meta-analysis of 13 trials and more than 91,000 participants found statin therapy carried a 9 percent increased relative risk of incident diabetes (odds ratio 1.09), which worked out to treating 255 people with a statin for 4 years to produce one additional case of diabetes.[34] A larger 2024 analysis of individual participant data from the same collaboration, covering 19 placebo-controlled trials and nearly 124,000 people, found low or moderate intensity statin therapy carried a 10 percent proportional increase in new-onset diabetes (1.3 percent of people affected per year versus 1.2 percent on placebo, a difference of one-tenth of one percentage point per year), while high-intensity statin therapy carried a substantially larger 36 percent proportional increase (4.8 percent per year versus 3.5 percent, a difference of 1.3 percentage points per year).[35] This is a genuine, dose-related effect, largest at the highest statin doses. It cuts the opposite direction from how it sometimes gets used in arguments: a drug that nudges blood sugar the wrong way is a reason the terrain-level work in the rest of this guide, insulin sensitivity, visceral fat, sleep, and nutrition, matters more, not less, alongside the medication rather than instead of it.

Second, muscle aches and pains are the single most common reason people say they want off a statin, and the honest answer is that trial data and real-world experience genuinely disagree with each other in a way that is not fully resolved. A 2019 American Heart Association scientific statement on statin safety found that across randomized, placebo-controlled trials, the difference in muscle symptoms between statin-treated and placebo-treated participants was under 1 percent, and under 0.1 percent for symptoms severe enough to stop the drug, while noting that roughly 10 percent of patients discontinue a statin in ordinary clinical practice, overwhelmingly citing muscle complaints as the reason.[36] A 2021 trial published in the BMJ tested this directly: 151 patients who had previously stopped a statin because of muscle symptoms were rotated, blinded, through repeated statin and placebo periods, and found no measurable difference in muscle symptom scores between the statin periods and the placebo periods.[37] Both findings are true at once: trial-measured rates are low, real-world discontinuation rates are meaningfully higher, and researchers do not fully agree on how much of that gap reflects a true drug effect trials underestimate versus expectation and attention effects that grow once someone knows what they are taking. If you have had muscle symptoms on a statin, that experience is real and worth a specific conversation with your prescriber, including whether a different statin, a different dose, or a supervised trial off and back on makes sense. It is a decision for that conversation, not one to make alone.

Put plainly, here is what is settled and what is not. Settled, and not seriously contested among researchers: LDL's causal role in cardiovascular disease, the strength of secondary-prevention benefit for people who have already had a cardiac event, and the diabetes-risk effect described above. Genuinely contested, or still being actively debated: the true size of absolute benefit for primary prevention in lower-risk people, the real-world rate of muscle symptoms outside of controlled trials, and how much all-cause mortality improves specifically in low-risk primary-prevention groups as opposed to higher-risk populations overall.

Everything else this guide covers, insulin sensitivity, inflammation, liver health, blood pressure drivers beyond salt, sleep, nutrition, and movement, is complementary work either way. It addresses the underlying terrain that produced the numbers your prescriber is treating, alongside that conversation, not instead of it.

Where this leaves your prescription. None of the above is a reason to distrust your prescriber or the medication itself, and it is also not a reason to treat "everyone should be on a statin" as beyond discussion. If you have documented cardiovascular disease, a prior heart attack or stroke, or a stent or bypass, the evidence for staying on your statin exactly as prescribed is strong and not seriously contested. This guide has nothing to add to that conversation except to keep taking it. If you have not had one of those events and are looking at a rising number for the first time, the honest, evidence-based position is that this is a real, individual decision with a genuinely small absolute benefit at the lower end of risk, one worth a specific conversation with your prescriber rather than an automatic yes or an automatic no. Either way, any decision about starting, stopping, changing, or adjusting a statin, a blood pressure medication, or a blood thinner belongs to you and your prescribing physician, based on your actual risk category and follow-up numbers, never to a supplement company, a podcast, or this guide.

If your functional-medicine work moves your numbers meaningfully over time (which is genuinely possible and is the entire point of addressing the terrain), that same rule applies: any change to dosing or discontinuation is a conversation to have with your prescribing physician, never a decision to make unilaterally or based on how you feel. Bring your ApoB, Lp(a), and triglyceride-to-HDL numbers from Key 2 to that conversation, along with where you fall on the primary-versus-secondary-prevention picture above. They give your prescriber a sharper, more complete basis for that specific decision, not a reason to second-guess what has already been prescribed.

The same logic applies to blood pressure medications and blood thinners. ACE inhibitors, ARBs, beta blockers, calcium channel blockers, diuretics, and anticoagulants each treat a specific, documented mechanism, and stopping or reducing any of them without your prescriber's involvement carries real risk, including the stroke and cardiac emergencies described earlier in this guide.

KEY 4

Inflammation and the Health of Your Blood Vessel Lining

Where plaque actually starts. Plaque does not build up in a healthy, smooth artery. It builds up where the single layer of cells lining the inside of your blood vessels, the endothelium, has already been disrupted. That disruption is called endothelial dysfunction, and it typically develops years before any cholesterol number or blood pressure reading looks abnormal.

How the vessel lining breaks down. A healthy endothelium produces nitric oxide, a signaling molecule that keeps blood vessels relaxed, discourages clotting, and calms inflammation locally. Sustained exposure to the risk factors this guide covers (insulin resistance, elevated blood pressure, oxidative stress, smoking) reduces nitric oxide availability, and endothelial dysfunction is now recognized as an early, reversible precursor of atherosclerosis, the pathophysiological link between early risk factors and the plaque that eventually causes heart attacks and strokes.[12] That word, reversible, describes this specific, early cellular process, not a promise about an existing diagnosis.

High-sensitivity C-reactive protein, or hs-CRP, is a blood marker of that inflammatory process, produced by the liver in response to inflammatory signaling elsewhere in the body. It adds prognostic information on cardiovascular risk that is comparable in strength to blood pressure or cholesterol themselves, and current risk categories treat a level under 1 mg/L as lower risk, 1 to 3 mg/L as average, and above 3 mg/L as higher relative risk.[11] In a 30-year study following more than 27,000 initially healthy women, hs-CRP, LDL cholesterol, and Lp(a) each independently predicted long-term cardiovascular events, with hs-CRP carrying the single strongest association of the three.[13] Inflammation is not a side note to the cholesterol story. In this data, it was the most powerful of the three markers measured.

What to ask your physician for. hs-CRP is a standard, inexpensive blood test your physician can order, and it is worth asking for if it has not already been checked. Read it for what it is: a nonspecific screening marker. It tells you an inflammatory load exists somewhere. It does not tell you where that load is coming from, and it confirms nothing on its own. A recent infection, an injury, a flaring joint, or a dental problem will all raise it, which is why it is interpreted alongside your other numbers and repeated when you are not acutely ill. What lowers it tends to be the same terrain-level work covered throughout this guide: addressing insulin resistance, visceral fat, sleep, and the nutrition and movement patterns in Key 9, all of which independently calm this inflammatory signal.

KEY 5

Your Liver Is a Metabolic Organ, Not Just an Alcohol Story

An outdated assumption. For decades, a fatty liver on imaging was assumed to be an alcohol problem, and people who did not drink heavily were often confused or dismissive when the finding showed up on their own scan. That assumption is outdated. In most cases in the United States today, a fatty liver has nothing to do with alcohol at all.

The name change, and why it happened. The condition long called non-alcoholic fatty liver disease (NAFLD) was formally renamed in 2023 by a multi-society consensus of liver specialists from 56 countries. The new name is metabolic dysfunction-associated steatotic liver disease, or MASLD, and the new diagnostic definition requires the presence of at least one of five cardiometabolic risk factors (excess weight, elevated blood pressure, elevated triglycerides, low HDL, or elevated blood sugar), replacing the old approach of diagnosing it only by ruling out alcohol.[14] The name change reflects what the field already understood mechanistically: this is fundamentally the liver's expression of the same insulin resistance driving Key 1, not primarily a toxin exposure story. Chronically elevated insulin drives the liver to convert excess circulating fuel into fat and store it directly inside liver cells, and over time that stored fat drives local inflammation and, in a meaningful share of cases, progressive scarring (fibrosis).

This matters well beyond the liver itself. A meta-analysis of 36 studies following more than 5.8 million people found that having a fatty liver was associated with a 45 percent higher rate of fatal or non-fatal cardiovascular events compared with not having one, a risk that climbed further, to roughly two and a half times higher, in people with more advanced fibrosis, independent of age, sex, adiposity, diabetes, and other standard cardiometabolic risk factors.[15] Practically, that means a fatty liver finding on an ultrasound is not an isolated footnote. It is a signal worth treating as seriously as an abnormal lipid panel, because in this data, it carried independent cardiovascular weight of its own.

What actually helps the liver. The liver responds well to the same terrain-level changes that improve insulin sensitivity elsewhere: reduced intake of refined carbohydrate and added sugar, regular movement, and, where relevant, weight change through the lifestyle patterns covered in Key 9. Elevated liver enzymes (ALT, AST, and GGT) on a standard metabolic panel are inexpensive early clues worth tracking over time rather than dismissing as a minor lab flag, and they belong on the worksheet at the end of this guide alongside your lipid and blood pressure numbers.

KEY 6

Blood Pressure Is Bigger Than Salt

More than one lever. “Cut your salt” is the most common piece of advice given for high blood pressure, and for some people it genuinely helps. But it is far from the whole story, and treating sodium as the only lever leaves several other well-documented drivers unaddressed.

What raises pressure besides sodium. As covered in Key 1, chronically elevated insulin itself promotes sodium retention in the kidney and increases sympathetic nervous system activity, both of which raise blood pressure independent of how much salt is actually on your plate. Two minerals matter more than most people realize: a dose-response meta-analysis of potassium supplementation trials found a real blood pressure-lowering effect, strongest in people who already have hypertension and in those with high sodium intake, though the relationship is U-shaped rather than “more is always better,” with the benefit leveling off and reversing at very high intakes, which is exactly why potassium status is worth assessing with your physician rather than self-supplementing at high doses, particularly if you take an ACE inhibitor, ARB, or potassium-sparing diuretic, or have any kidney impairment.[16] Magnesium supplementation shows a similar pattern in a 2025 meta-analysis of 38 randomized trials: a modest average reduction in blood pressure across all participants, but a substantially larger reduction, roughly seven to eight points systolic, specifically in people already being treated for hypertension or who had low magnesium status to begin with.[17] Read together, both minerals appear to matter most for the people who need the most help, not as a blanket fix for everyone.

Two other drivers deserve more attention than they usually get. Obstructive sleep apnea, repeated pauses in breathing during sleep that fragment rest and spike stress hormones overnight, is strongly linked to resistant hypertension, the kind that does not respond well to standard medication combinations. A meta-analysis found people with obstructive sleep apnea were roughly three to four times more likely to have resistant hypertension than people without it.[18] If your blood pressure is difficult to control despite adherence to medication, snoring, witnessed pauses in breathing, or unrefreshing sleep are worth mentioning to your physician as a possible undiagnosed contributor. Chronic psychological stress also shows a real, if still-debated, association with hypertension risk, with one meta-analysis finding people under chronic psychosocial stress were roughly two and a half times more likely to have hypertension, while the authors themselves noted the underlying studies vary enough in how they define and measure stress that more research is needed to firm up the relationship.[19] Kidney function matters too, in both directions: the kidneys and the blood pressure system regulate each other through the renin-angiotensin-aldosterone pathway, so impaired kidney filtration can itself drive blood pressure up, which is part of why a basic metabolic panel checking kidney markers belongs alongside a blood pressure log.

The DASH evidence, and what to ask for. Beyond sodium reduction, the DASH dietary pattern (rich in fruits, vegetables, and low-fat dairy, with reduced saturated fat) has one of the strongest evidence bases of any dietary blood pressure intervention: the original randomized trial found it lowered systolic and diastolic blood pressure by 5.5 and 3.0 points respectively compared with a typical diet, with a substantially larger 11.4 and 5.5 point reduction in the subgroup who already had hypertension.[20] Ask your physician about checking your magnesium and potassium status, mention sleep quality and snoring at your next visit, and consider a home blood pressure log (the worksheet at the end of this guide includes one) so patterns show up before your next appointment rather than only in a single office reading.

KEY 7

Visceral Fat, Why Where You Store It Matters

Same weight, different risk. Two people can carry the same total body weight and the same BMI and have meaningfully different cardiometabolic risk, because it is not just how much fat you carry, it is where you carry it.

Why location changes the risk. Fat stored around and inside the abdominal organs, visceral fat, behaves entirely differently from fat stored just under the skin. A comprehensive review of the physiology describes visceral fat as part of a broader pattern, including dysfunctional fat cell expansion and fat stored in places it does not belong (the liver and muscle), that is closely tied to the clustering of cardiometabolic risk factors covered throughout this guide: elevated triglycerides, increased free fatty acid release into the bloodstream, release of inflammatory signaling molecules directly from fat tissue, liver insulin resistance, small dense LDL particles, and reduced HDL.[21] In practical terms, visceral fat is not passive storage. It is an active, inflammatory tissue that manufactures many of the same downstream problems described in Keys 1 through 6.

This is also why waist circumference and waist-to-hip ratio carry real, independently documented predictive value. The INTERHEART study, a standardized case-control study spanning 52 countries and more than 27,000 participants, found abdominal obesity (measured by waist-to-hip ratio) was one of just nine modifiable factors that together accounted for roughly 90 to 94 percent of the population-level risk of a first heart attack worldwide, alongside smoking, an elevated ApoB-to-ApoA1 ratio (the same particle-count logic from Key 2, which on its own accounted for the second-largest share of risk after smoking), high blood pressure, diabetes, psychosocial stress, and diet and activity patterns.[22] That is a global, cross-cultural dataset confirming that where you store fat, not just how much you weigh, is one of the small handful of factors that explain most heart attack risk worldwide.

What to track, and why it moves first. A simple waist circumference measurement, taken at the navel, adds real information beyond the number on the scale or your BMI, and it is worth tracking over time on the worksheet at the end of this guide. The encouraging part of this research is that visceral fat responds preferentially to the same lifestyle interventions in Key 9. When people lose weight through nutrition and movement changes, visceral fat tends to mobilize first, ahead of fat stored elsewhere.

KEY 8

Sleep and Circadian Rhythm, the Underrated Driver

Treated as a nicety, not a driver. Sleep is treated as a lifestyle nicety in most cardiovascular conversations, something to improve "if you have time." The evidence does not support that ranking. Sleep duration and timing function as a direct cardiometabolic input, not an afterthought.

What the data actually shows. A systematic review and meta-analysis following nearly 475,000 people found that short sleep duration was associated with a 48 percent higher risk of developing or dying from coronary heart disease and a 15 percent higher stroke risk, and, somewhat counterintuitively, long sleep duration (generally more than eight or nine hours) was also associated with increased risk across coronary heart disease, stroke, and total cardiovascular disease, suggesting the relationship is U-shaped rather than "more sleep is always better."[23] Shift work adds a separate, related problem: circadian misalignment, the mismatch between your internal biological clock and your actual sleep and eating schedule, is itself an independent risk factor for overweight, obesity, type 2 diabetes, elevated blood pressure, and metabolic syndrome, distinct from sleep duration alone.[24] The mechanism runs back through the same insulin pathway covered in Key 1: circadian disruption impairs glucose regulation and insulin sensitivity even when total sleep time looks adequate on paper.

What to prioritize. Consistent sleep and wake times, even on weekends, appear to matter as much as total sleep duration. If your schedule includes rotating or overnight shifts, that is a genuine, independent cardiometabolic risk factor worth naming explicitly to your physician, not just a scheduling inconvenience. The worksheet at the end of this guide includes a simple place to log typical sleep hours alongside your other numbers, since patterns over weeks tell you more than any single night.

KEY 9

Nutrition and Movement, What Actually Has Evidence Behind It

Noisy advice, specific answers. Nutrition advice for cardiometabolic risk is noisy, contradictory, and often reduced to “eat less, move more” without much specificity. Several interventions in this space have real, randomized evidence behind them, and it is worth knowing which ones.

What the trials actually tested. A Mediterranean-pattern diet, emphasizing olive oil, vegetables, legumes, fish, and nuts, has been tested directly against a standard low-fat diet advice approach in a large Spanish randomized trial of over 7,400 adults at high cardiovascular risk. After correcting for early protocol irregularities at some study sites and reanalyzing the full dataset, the corrected results still showed roughly a 30 percent reduction in major cardiovascular events (heart attack, stroke, and cardiovascular death combined) in the groups assigned to a Mediterranean diet supplemented with either extra-virgin olive oil or mixed nuts, compared with the reduced-fat control diet.[25] Since this guide reports funding everywhere it matters: that trial was funded by Spain’s public health research institute, with the olive oil and nuts supplied as donations from olive oil and nut industry groups which, per the published report, had no role in the design or analysis of the study.[25] On the movement side, a randomized trial directly measuring insulin sensitivity found that a single session of either moderate-intensity continuous exercise or high-intensity interval training produced roughly a 20 percent improvement in insulin sensitivity the very next day compared with the pre-training baseline, with both exercise styles working equally well, though the benefit faded within about four days without another session, which is exactly why consistency matters more than any single “best” workout format.[26] On the fiber side, a meta-analysis of 28 randomized trials found that at least 3 grams daily of oat beta-glucan, a soluble fiber, lowered LDL cholesterol by an average of about 0.25 mmol/L (roughly 10 mg/dL) without changing HDL or triglycerides.[27] None of these numbers are dramatic in isolation. Together, sustained over time, they represent a genuinely evidence-based lever most people never hear quantified.

The three concrete levers. None of this requires a rigid or extreme diet. The practical takeaway is specific rather than general: a Mediterranean-style eating pattern most days, movement most days of the week (even moderate-intensity movement shows the same insulin-sensitizing effect as harder interval training), and a deliberate daily source of soluble fiber (oats, legumes, and psyllium are the best-studied sources) are three concrete, evidence-supported levers, not vague lifestyle suggestions. These changes work on the same insulin resistance root cause from Key 1, which is why they tend to move cholesterol, blood pressure, and liver fat together rather than requiring three separate diets for three separate numbers.

Your Staged Path Forward

Everything above describes the problem and the general categories of solution. Actually working through it in your own case benefits from a structured, staged approach rather than trying to change everything at once.

Stage 1: Get the full picture. Beyond your standard lipid panel and a single blood pressure reading, ask your physician about ApoB, a one-time Lp(a), hs-CRP, fasting insulin, homocysteine, and liver enzymes (ALT, AST, GGT), and start logging blood pressure at home using the worksheet below. You cannot address what you have not measured.

Stage 2: Address the foundational terrain. For a defined trial period, commonly eight to twelve weeks, focus on the nutrition pattern, movement, sleep consistency, and stress covered in Keys 6 through 9, while continuing every prescribed medication exactly as directed. This stage is about the modifiable terrain, not about judging or replacing what your prescriber already started.

Stage 3: Reassess and personalize. Repeat the labs and blood pressure log from Stage 1, and bring the results to your prescribing physician. This is the point where genuinely individualized decisions get made, including whether targeted nutrient support (magnesium, potassium, fiber) makes sense for your specific numbers, whether a sleep study is warranted, and whether any medication adjustment is appropriate, a decision that belongs to your prescriber based on your actual data.

Stage 4: Maintain and recheck periodically. Cardiometabolic risk is a long arc, not a single fix. Periodic rechecking, roughly every three to six months initially, keeps the picture current as your terrain continues to change.

Your Numbers Worksheet

Tracking your own numbers over time is one of the most genuinely useful things you can do with this information. Bring this log to every appointment.

Blood pressure and vitals

Date	Systolic	Diastolic	Resting heart rate	Weight	Waist circumference

Labs (repeat roughly every 3 to 6 months, or as your physician directs)

Draw date	ApoB	LDL-C	Lp(a) (once is usually enough)	Triglycerides	HDL	TG/HDL ratio	hs-CRP	Homocysteine	Fasting glucose	HbA1c	Fasting insulin	ALT / AST / GGT

Fasting insulin sits in that row next to glucose and HbA1c on purpose. Insulin climbs for years to hold glucose in range, so a blood sugar workup that measures only the two downstream numbers finds the problem late. Ask for all three together.

Sleep and lifestyle notes

Week of	Average sleep hours	Shift work this week?	Movement days	Notable stress

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a free consultation with me to review your specific numbers: what your labs and blood pressure actually show, which drivers from this guide look most relevant in your case, and what a personalized, prescriber-coordinated plan could look like. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at 919-662-0520.

If you would rather start on your own, start tracking.

The numbers worksheet earlier in this guide works on paper. The digital version does the same job on your phone, keeps your history in one place, and lets you watch the numbers that actually move: home blood pressure readings over weeks rather than one anxious reading in an exam room, and how they respond as you change things. Bringing that history to your next lab review makes it dramatically more useful than a single point in time.

Start your free tracker at rootcauseunlocked.com/tracker.

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Every citation below was independently retrieved and confirmed against its live PubMed record (esearch plus esummary and, for every citation, full abstract confirmation via efetch) before use. None are drawn on trust from any prior document, including the practice's own internal protocol guides. Full verification detail, including several claims investigated and left uncited or unused, is in the companion file Heart-Health-Citation-Ledger.md.

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This guide is provided for general educational purposes and does not constitute individualized medical advice, diagnosis, or treatment. It does not replace regular cardiovascular screening or your physician's evaluation. Always consult your physician before starting any new supplement or exercise program, particularly if you are pregnant, nursing, have kidney disease, or are taking prescription medications including statins, blood pressure medications, or blood thinners. Never stop, skip, or change a prescribed medication based on this guide. If you have diagnosed heart disease, kidney disease, or have had a cardiac event, this guide is educational background only and does not replace your cardiologist's or nephrologist's direction of your care. If you are experiencing any symptoms described in the emergency section of this guide, call 911 or seek emergency care immediately.

From Understanding to Doing

Everything up to this point was about understanding: what is happening in your body, why it may be happening, and which directions are worth investigating. That is the half a book can do well.

What follows is the other half, and it is a different kind of document on purpose. Three guides, written to be used rather than read straight through. The first tells you which lab panels answer which question and what the numbers mean, and it includes a request list you can hand to a clinician. The second covers what to eat and, more usefully, how to test a dietary change on yourself so you end up with an answer instead of a habit. The third walks through what has actually been studied, at what dose, and names the products whose trials came back negative.

Each one stands alone, so use them in whatever order matches what you are stuck on. If you are not sure where to begin, begin with the lab guide, because knowing what is driving your symptoms changes what you should do about the other two.

A note on repetition. A few points appear in both halves. That is deliberate. The ones repeated are the ones where getting it wrong causes harm, and they are worth meeting twice.

Before anything else: when to stop reading and call 911

Call 911 now if you have:

- Chest pain, pressure, tightness, or squeezing, especially with sweating, nausea, breathlessness, or pain spreading to the jaw, neck, back, or arm. **Do not drive yourself.**
- Sudden weakness or numbness on one side, face drooping, trouble speaking or understanding, sudden vision loss, or sudden severe unexplained headache. Note the time symptoms started, because treatment decisions depend on it.
- Sudden severe breathlessness, or coughing pink frothy sputum.
- A tearing or ripping pain in the chest or back.

Women, and people with diabetes, more often present without classic crushing chest pain. Unexplained breathlessness, unusual fatigue, nausea, or jaw and back discomfort can be the presentation. If something feels badly wrong and new, that is enough reason to be seen.

This section is first on purpose. Everything else in this kit is for a calm day.

Read this before you read anything about statins

Key 3 of *Unlock Your Heart Health* is titled around what the statin evidence actually shows. This kit takes the same position, and I want it clear from the first page:

Nothing in these guides is a reason to stop a prescription on your own. Stopping a cardiovascular medication on the strength of a wellness document is one of the few things in this entire product line that could genuinely harm you.

What the supplement guide does instead is give you the whole picture, including the parts both sides leave out: that industry-funded drug research is measurably more favourable to the sponsor, that **in true primary prevention lipid-lowering therapy has not been shown to reduce mortality**, and that the muscle-symptom objection does not survive blinded testing. Those three things are all true at once, and knowing all three is what lets you have a real conversation instead of a one-sided one.

The nine keys, and what tests each one

Key	What it means	How you find it
1. Insulin resistance	The shared root of cholesterol, blood pressure, and fatty liver	Fasting insulin, glucose, HbA1c, triglyceride to HDL ratio
2. Beyond the lipid panel	What a standard panel leaves out	**ApoB and Lp(a).** The highest-yield additions here
3. Statin evidence	What it shows and does not	Covered in the supplement guide
4. Inflammation	The blood vessel lining	hs-CRP
5. Liver	A metabolic organ, not just an alcohol story	ALT, AST, GGT, imaging, fibrosis scoring
6. Blood pressure	Bigger than salt	A validated home monitor. Not a blood test
7. Visceral fat	Where you store it matters	A tape measure. Not a blood test
8. Sleep	The underrated driver	Sleep apnea assessment, sleep log
9. Nutrition and movement	What has evidence behind it	The diet guide

Three of the nine are not blood tests at all, and one of those, a tape measure around your waist, may tell you more than most of the panel.

Key 2: the two tests your standard panel probably left out

ApoB

What it is. Every atherogenic particle carries exactly one apolipoprotein B molecule. So ApoB counts the particles, while LDL cholesterol measures the cholesterol carried inside them. Those are different, and they come apart in a specific group of people.

Why it matters. A review comparing ApoB with non-HDL cholesterol as a treatment target concluded that ApoB is preferred as a secondary target in people with mild-to-moderate high triglycerides (175 to 880 mg/dL), diabetes, obesity or metabolic syndrome, or very low LDL cholesterol under 70 mg/dL, and that where ApoB is unavailable, non-HDL cholesterol should be used to supplement LDL cholesterol (PMID 32562186).

Read the list of who that applies to. Diabetes, obesity, metabolic syndrome, high triglycerides. That is precisely the population Key 1 of this book is about. If insulin resistance is your picture, your LDL cholesterol can look reassuring while your particle count does not.

What to ask for. ApoB. If your lab does not offer it, ask for **non-HDL cholesterol**, which is simply total cholesterol minus HDL and can be calculated from the panel you already have.

Lipoprotein(a)

What it is. A genetically determined lipoprotein, largely set by inheritance and barely moved by diet or lifestyle.

Why it matters. Lp(a) is an independent risk factor for atherosclerotic cardiovascular disease including acute coronary syndrome, peripheral arterial disease and stroke, and for calcific aortic stenosis, with an estimated 20% of the population affected (PMID 41544761).

The practical points.

- **Measure it once in your life.** It is largely genetic and does not need repeating.
- **A high result does not mean you failed at lifestyle.** It means your baseline risk is higher and everything else you can control matters more.
- **It has family implications.** If yours is high, first-degree relatives are worth testing.
- Targeted therapies are in development, and current management focuses on controlling everything else more aggressively.

Key 1: insulin resistance, the shared root

The ebook's argument is that cholesterol, blood pressure and fatty liver share one engine. Test the engine.

What to ask for: fasting insulin, fasting glucose, and HbA1c.

And one ratio you can calculate yourself right now, from a panel you probably already have: **triglycerides divided by HDL**. It costs nothing, it is not a validated diagnostic threshold, and it is a useful pointer toward the insulin-resistant pattern when the ratio is high. Bring it to your appointment as a question rather than a conclusion.

Why fasting insulin matters. Glucose stays normal for years while insulin climbs to hold it there. Measuring only glucose means finding the problem after the compensation fails, which is exactly the window you want to act inside.

Key 4: inflammation

hs-CRP, the high-sensitivity version, not standard CRP. Standard CRP is built to detect infection-level inflammation; the high-sensitivity assay reads the low range that matters for vascular risk.

One recent paper describes bundled measurement of LDL cholesterol, Lp(a), and high-sensitivity CRP across the risk spectrum, which is a reasonable way to think about ordering all three together rather than piecemeal (PMID 41509673).

How to read it. A single raised hs-CRP after a cold means nothing. Repeat it when you are well before drawing conclusions.

Homocysteine: the marker that needs the most honest handling in this kit

What it is. An amino acid your body produces and clears in the methylation cycle, a set of reactions that runs on B12, folate, and B6. It backs up in the blood when one of those vitamins runs short, when kidney or thyroid function is off, or when a common MTHFR gene variant slows the recycling enzyme.

What an elevated result actually means. Treat it as a signal to investigate B12, folate, B6, thyroid, and kidney status, not as a cardiac target proven to change outcomes on its own. That distinction is the whole game with this marker, and a lot of wellness content gets it backwards.

The honest evidence, in three sentences. Observationally, lower homocysteine tracks with modestly lower heart disease and stroke risk, "at most a modest independent predictor" in the pooled data (PMID 12387654). Genetically, lifelong moderate elevation showed little or no effect on coronary disease (odds ratio 1.02) once 19 unpublished datasets were included, with the published literature's positive result attributed to publication bias (PMID 22363213). And in randomized trials, lowering it with folic acid cut stroke by about 10 percent, mostly in folate-deficient populations, while coronary disease did not move (PMID 27528407).

Notice which direction that bias ran. The homocysteine literature skewed toward the position wellness culture favors, exactly as industry funding skews drug literature toward the sponsor's position. This kit applies the same test to both, and reports what survives it.

Which range frame you are read against matters. At Root Health I read against a functional target under 7 umol/L. Standard laboratory ranges run to 10 or 15 depending on the lab. A result above roughly 10 is worth the workup conversation either way.

If it comes back elevated, the follow-up panel is: B12, folate, B6, TSH, and kidney function (creatinine and eGFR). MTHFR genotyping is optional; it changes which vitamin form makes sense, not the conclusion.

One boundary. Everything above concerns moderate elevation. Homocystinuria, the rare inherited disorder where levels run several times normal, is a genuinely dangerous medical condition, and none of the reassurance above applies to it.

Key 5: your liver

Fatty liver is now usually called metabolic dysfunction-associated steatotic liver disease, or MASLD, and the name change is the point: it is a metabolic condition, not primarily an alcohol story.

What to ask for. ALT, AST, and GGT to start. A normal ALT does not exclude fatty liver, which is the most common misunderstanding here. If there is suspicion, imaging and a non-invasive fibrosis score are the next step, and that is a physician's call.

Why it is worth finding. A systematic review and network meta-analysis of randomized trials found sustained weight loss associated with histological improvement in MASLD in people with obesity, while being explicit that the limited evidence base and sparse network preclude causal inference and that the findings should be treated as exploratory and hypothesis-generating (PMID 41804193).

That is a fair summary of where the field is: weight loss is the lever, and the trial base is thinner than the confidence with which it is usually stated.

Keys 6 and 7: the two measurements that are not blood tests

Blood pressure

Buy a validated upper-arm home monitor. Wrist cuffs are less reliable. A single reading in a clinic is a poor measure of anything, and home readings across a week are far better.

How to take it properly, because most home readings are taken wrong: sitting, back supported, feet flat, arm supported at heart height, no talking, after five minutes of quiet, and not within thirty minutes of caffeine, exercise or smoking. Two readings a minute apart, morning and evening, for seven days.

Your waist

A tape measure around your waist at the level of the navel tells you about visceral fat, which is Key 7 and is more metabolically active than fat elsewhere. Measured over a year, it is a better progress signal than the number on the scale, because it moves when body composition changes even if weight does not.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose. **Lipid targets are deliberately absent from this table**, because the right LDL or ApoB target depends on your overall risk, your history, and whether you are in primary or secondary prevention, and a one-size number would mislead. That is a conversation with your physician using a risk calculator.

Marker	Units	Functional target	Approximate standard range
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
hs-CRP	mg/L	under 0.5	under 1.0
Ferritin	ng/mL	50 to 150	25 to 300
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Homocysteine	umol/L	under 7.0	under 10 to 15, varies by lab
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5

Reading your results as a pattern

Pattern one: the metabolic cluster. High triglycerides, low HDL, creeping fasting insulin, raised ALT, expanding waist, blood pressure drifting up. That is one problem wearing five labels, and Key 1 is the answer.

Pattern two: reassuring LDL, unreassuring ApoB. Common in exactly the group named in the ApoB review above. The particle count is the number to act on.

Pattern three: everything good except Lp(a). Genetic, not your fault, and it raises the value of controlling everything you can control.

Pattern four: normal panel, real symptoms. If you have exertional chest discomfort or breathlessness, a normal lipid panel does not clear you. That is a cardiology question, not a supplement question.

The conversation with your doctor

A request list you can bring

- Standard lipid panel, plus **ApoB**, or non-HDL cholesterol if ApoB is unavailable
- **Lp(a), once in your life**
- Fasting insulin, fasting glucose, HbA1c
- hs-CRP
- Homocysteine, read as a B-vitamin, thyroid, and kidney workup trigger, not a cardiac target
- ALT, AST, GGT
- A cardiovascular risk estimate using a validated calculator, with your numbers in it
- Sleep apnea assessment if you snore or wake unrefreshed

The sentence that helps

"My LDL looks fine but I have metabolic syndrome. Could we check ApoB, since I understand LDL cholesterol can underestimate particle count in that setting?" Specific, accurate, and it opens the right conversation.

What these labs cannot tell you

- **Whether you have plaque.** Lipids estimate risk. Imaging such as a coronary calcium score answers a different and more direct question, and is worth discussing with your physician.
- **Your risk on their own.** Risk is calculated from a combination, not read off a single line.
- **Whether you need a statin.** That is a risk-based decision with your physician, covered in the supplement guide.

When to seek clinical review

Emergency, repeated from the top: chest pain or pressure, one-sided weakness or speech trouble, sudden severe breathlessness, tearing chest or back pain. Call 911.

Soon: blood pressure repeatedly at or above 140/90 at home, chest discomfort on exertion that eases with rest, new breathlessness on stairs you used to manage, swelling in both ankles, or an ALT that stays raised.

Your next step

The companion guides do the rest: **The Heart and Metabolic Diet Guide** covers the eating pattern with the best blood pressure and metabolic evidence, and **The Heart and Metabolic Supplement Guide** covers what genuinely helps alongside your medications, including the honest read on red yeast rice.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=heart-labs***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=heart-labs***

If you would rather start on your own, start with home blood pressure readings and a waist measurement. Those two cost almost nothing and they move:

> ***Free tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=heart-labs***

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All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Heart-Kit-Citation-Ledger.md`.

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Educational content only. Not medical advice. Not a diagnosis. Not a reason to stop or refuse any cardiovascular medication. Chest pain is a medical emergency until proven otherwise.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Call 911 for chest pain or pressure, especially with sweating, nausea, breathlessness, or pain into the jaw, neck, back or arm; for sudden one-sided weakness, face droop, or trouble speaking; for sudden severe breathlessness; or for tearing chest or back pain. Do not drive yourself.

Nothing here is a reason to stop or refuse a statin, a blood pressure medication, or metformin.

If you have kidney disease, heart failure, or take an ACE inhibitor, ARB, or potassium-sparing diuretic, **talk with your clinician before increasing potassium or using a potassium-based salt substitute.** That caution appears again where it matters below, and it is the one item in this guide that can hurt the wrong person.

The one idea underneath all three conditions

Key 1 of *Unlock Your Heart Health* argues that high cholesterol, high blood pressure, and fatty liver share a root: insulin resistance. That is why this is one diet guide rather than three.

It also means the food work is not primarily about avoiding dietary cholesterol or counting fat grams. It is about the pattern that improves insulin sensitivity, blood pressure, and liver fat at the same time. Those move together, and so do the interventions.

Part one: what actually lowers blood pressure

This is the section with the best evidence-to-effort ratio in the whole kit.

What a network meta-analysis found. In adults ranging from prehypertension to established hypertension, comprehensive lifestyle modification, aerobic exercise, isometric training, **low-sodium and high-potassium salt**, salt restriction, breathing-control, meditation, and a low-calorie diet all showed clear effects on blood pressure reduction (PMID 32975166).

Note what is on that list. Not just salt restriction. Breathing and meditation are on it too, which is unusual for a cardiology paper and worth taking seriously.

The sodium and potassium point, which is better than “eat less salt”

A systematic review and meta-analysis found that a lower 24-hour urinary sodium to potassium ratio is associated with lower blood pressure in adults, and concluded that dietary strategies raising potassium while lowering sodium would be beneficial for blood pressure (PMID 34117485).

That reframes the task usefully. It is a ratio. You can improve it from either end, and adding potassium-rich food is easier to sustain than joylessly cutting salt.

Potassium-rich foods: potatoes with skin, beans and lentils, leafy greens, tomato products, avocado, banana, citrus, yogurt, salmon.

The safety line, repeated because it matters: if you have kidney disease or heart failure, or take an ACE inhibitor, ARB, or potassium-sparing diuretic, **do not increase potassium or use a potassium salt substitute without your clinician**. In those situations high potassium is genuinely dangerous.

Where the sodium actually is. Not the shaker. Bread, deli meat, cheese, sauces, soups, and restaurant food carry most of it. Cooking at home more often does more than removing the shaker ever will.

The DASH pattern, with the honest evidence grade

DASH is the most recommended eating pattern for blood pressure. A 2026 systematic review and meta-analysis comparing DASH against alternative structured dietary interventions found benefits on blood pressure and selected cardiometabolic parameters, while stating plainly that **the overall certainty of evidence was low with substantial heterogeneity**, and that findings should be interpreted cautiously (PMID 42492868).

How to read that. DASH is a sensible, well-tested pattern. It is not clearly superior to other structured healthy patterns, and the comparative evidence is weaker than its reputation. So if you would rather eat a Mediterranean pattern, that is a reasonable choice, and adherence matters more than which structured pattern you pick.

The DASH shape, plainly: vegetables and fruit at most meals, whole grains, beans, nuts and seeds, low-fat dairy if you eat dairy, fish and poultry over red and processed meat, and less added sugar and sodium.

Part two: eating for insulin, which is the shared engine

The same five practices I use in every insulin-driven picture, because the engine is the same one:

1. **Protein at every meal, starting with breakfast.**
2. **Never eat carbohydrate naked.** Pair it with protein, fat, or fiber.
3. **Fiber at every meal**, and see the soluble fiber note below.
4. **Cut liquid sugar first.** Sweetened drinks and juice.
5. **Walk after meals**, ten to fifteen minutes.

Sourcing note. Those five are standard clinical nutrition for insulin resistance rather than findings from a cardiovascular outcome trial, and no citations are attached to them for that reason. The blood pressure and liver claims in this guide are cited.

Soluble fiber deserves a specific mention because it does two jobs at once here: it blunts the post-meal glucose rise and it lowers LDL cholesterol modestly and reliably. Oats, barley, beans, lentils, psyllium, apples, and citrus. Build up gradually with plenty of water.

Part three: the liver, which is Key 5

Fatty liver, now usually called metabolic dysfunction-associated steatotic liver disease, is a metabolic condition rather than primarily an alcohol story.

Weight is the lever. A systematic review and network meta-analysis of randomized trials found sustained weight loss associated with histological improvement in MASLD in people with obesity, while being explicit that the limited evidence base and sparse network preclude causal inference, and that the findings are exploratory and hypothesis-generating (PMID 41804193).

Coffee, which is the pleasant surprise in this guide. An umbrella review plus systematic review and meta-analysis of coffee and non-alcoholic fatty liver disease found mixed results for preventing NAFLD in the general population, but **benefits of coffee consumption on liver fibrosis among patients who already have NAFLD** (PMID 34966282).

Read that split precisely: coffee is not established as preventing fatty liver, and among people who have it there is a signal on fibrosis. Black or minimally sweetened, since a coffee drink carrying 40 g of sugar is working against the rest of this guide.

Fructose deserves naming. Liquid fructose in particular, meaning sweetened drinks and juice, is handled largely by the liver. This is the same recommendation as cutting liquid sugar above, arriving from a different direction, which is usually a sign it is worth doing.

Alcohol. Even in the metabolic form of fatty liver, alcohol adds a second injury to a liver already under load. If you have fatty liver, reducing it is a direct intervention rather than general advice.

Part four: a week that works

Breakfast. Oats with nuts and berries. Eggs with vegetables and beans. Plain yogurt with fruit and seeds.

Lunch. Lentil or bean soup with wholegrain bread. Large salad with fish or chicken, olive oil, and a fiber-rich carbohydrate.

Dinner. Fish two or three times a week. Otherwise poultry, beans, or lentils, with two vegetables and a starch such as potato with the skin on, which is also a potassium win.

Snacks. Nuts, fruit, yogurt. Pair rather than eat carbohydrate alone.

Drinks. Water, coffee, tea. This is the single largest change most people can make and it hits blood pressure, insulin, and liver at once.

Cooking at home more often is the mechanism behind most of the sodium reduction above, and it does not require a recipe list.

What I deliberately left out, and why

- **Blanket dietary cholesterol restriction.** For most people, dietary cholesterol is a much smaller lever than the pattern as a whole. Eggs are not the problem in most diets.
- **Very low fat diets.** Long out of step with the evidence, and hard to sustain.
- **Ketogenic diets as the default.** They can improve triglycerides and liver fat, and their effect on LDL is variable and sometimes markedly adverse. If you want to try one with a raised LDL or ApoB, that needs monitoring and a physician, not a guide.
- **Grapefruit, if you take a statin or several blood pressure medications.** A genuine interaction, and worth checking against your specific prescription.
- **Potassium salt substitutes for everyone.** Genuinely useful for many people and genuinely dangerous in kidney disease, heart failure, and on certain medications. This is a clinician question.
- **Detox and cleanse protocols for the liver.** No outcome evidence, and the actual intervention with evidence is weight, alcohol, and sugar.

Telling you what to skip is part of what you paid for.

Running a fair experiment

Weeks 1 and 2, baseline. Home blood pressure twice daily, waist measurement, and the labs from the lab guide.

Weeks 3 to 10, the pattern. Sodium down and potassium up, protein at breakfast, soluble fiber daily, liquid sugar out, walk after meals.

Week 11, measure. Blood pressure responds within weeks. Lipids and liver enzymes need eight to twelve weeks, so time your retest accordingly.

Realistic expectations

What is realistic. Meaningful blood pressure change within four to eight weeks from the sodium, potassium, movement and weight work. Modest LDL reduction from soluble fiber. Real liver improvement over months with sustained weight loss.

What is not realistic. Diet alone normalizing a genetically high Lp(a), or replacing appropriate medication in someone at high calculated risk. Those are additive, not alternatives.

When to seek professional care

Emergency: chest pain or pressure, one-sided weakness or speech difficulty, sudden severe breathlessness, tearing chest or back pain. Call 911.

Soon: home blood pressure repeatedly at or above 140/90, chest discomfort on exertion, new breathlessness on stairs, swelling in both ankles, or a persistently raised ALT.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your numbers, your calculated risk, and your medication list.

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Read the companion documents alongside this one. **Understanding Your Heart and Metabolic Labs** covers ApoB and Lp(a), and **The Heart and Metabolic Supplement Guide** covers red yeast rice honestly, including what it actually contains.

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Call 911 for chest pain or pressure, especially with sweating, nausea, breathlessness, or pain into the jaw, neck, back or arm; for sudden one-sided weakness, face droop, or trouble speaking; for sudden severe breathlessness; or for tearing chest or back pain. Do not drive yourself.

Nothing in this guide is a reason to stop a prescription on your own. The statin section below gives you the honest evidence in both directions, including who funded it, so you can have a real conversation with your prescriber instead of a one-sided one.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

How to read “no evidence” in this guide

The statin section below explains how funding shapes research results in this field. The same economics shape what never gets studied at all: a large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back, which nobody can do with magnesium or an herb. So when this guide says “no evidence,” it will tell you which of three things it means: *tested and failed* (a real trial ran and the answer was no), *never properly tested* (nobody could afford the trial, so absence of evidence is not evidence of absence), or *tested small and won* (real but modest trials, usually on markers rather than outcomes, because markers are what a small budget can afford).

The asymmetry earns nutrients humility, not automatic credit: small nutrient trials carry their own publication bias toward positive findings, just as drug trials tilt toward their sponsor (PMID 28207928). One rule, applied in both directions: name the funding, name the evidence tier, and let you see both.

Statins: the honest version, including the parts the drug companies do not advertise

Most of the functional medicine world is statin-skeptical, and a lot of that skepticism is well founded. Some of it is not. Sorting the two is worth more to you than a slogan in either direction, so here is the whole picture.

First, the criticism that is completely valid: who paid for the research

Industry-funded drug research produces more favourable results than independently funded research, and this is a documented, quantified effect rather than a suspicion. A Cochrane methodology review found that sponsorship of drug and device studies by the manufacturing company leads to more favourable efficacy results and conclusions than sponsorship from other sources, and concluded that this reflects **an industry bias that cannot be explained by standard risk-of-bias assessments** (PMID 28207928).

That last clause matters enormously. It means the usual quality checks a reviewer applies do not catch it. So "this was a well-conducted randomized trial" is not a sufficient answer to "who funded it."

The large statin efficacy trials, the ones establishing cardiovascular benefit, were overwhelmingly manufacturer-funded. Anyone who tells you that fact is irrelevant is not being straight with you.

Second, the distinction that changes everything: primary versus secondary prevention

This is the single most important thing in this guide and it is routinely collapsed in both directions.

Secondary prevention means you have already had a heart attack, a stroke, or documented cardiovascular disease. The evidence for statins here is strong and consistent, and this is not seriously disputed.

True primary prevention means you have none of that and are being treated on the basis of estimated risk. A 2026 meta-analysis of randomized trials looked specifically at people **without** prior cardiovascular, vascular, or cerebrovascular disease, because earlier analyses had blurred the line by including people who already had disease. In that genuinely primary-prevention population, lipid-lowering therapy **did not reduce all-cause mortality or cardiovascular mortality**, while it did significantly reduce **nonfatal** cardiovascular events and cardiovascular hospitalizations. The authors state that additional research is needed to define the benefits in true primary prevention (PMID 41864911).

Read that carefully, because it is the crux of the whole argument. In true primary prevention, the honest claim is a reduction in nonfatal events, not a demonstrated reduction in dying. That is a real benefit and it is a materially smaller claim than most people are given. If you are a healthy 52-year-old with a moderately raised LDL and no disease, that finding is about you, and you are entitled to know it before you decide.

Third, the criticism that does not hold up: muscle symptoms

Here the skepticism runs ahead of the evidence, and I checked the funding on these specifically because it is a fair question to ask of anything I cite.

StatinWISE was a series of **200 N-of-1 randomized controlled trials** in primary care, in which individual patients alternated blinded between statin and placebo. It was **funded by the UK National Institute for Health Research and the Department of Health, and run from the London School of Hygiene and Tropical Medicine. Not industry.** Most participants who completed the trial decided to continue taking statins afterwards (PMID 33709907).

A separate systematic review and meta-analysis of blinded statin rechallenge, from a university family medicine department declaring no competing interests, found **no significant difference in mean myalgia or global symptom score between statin and placebo**, that only about **one-third** of people previously believed to be statin intolerant were genuinely unable to tolerate a statin on blinded rechallenge, and that **one-quarter were intolerant of placebo** (PMID 38128013).

This does not mean anyone's symptoms are imaginary. Symptoms are real and felt. It means that attributing them to the statin without blinding is unreliable, and that most people who believe they cannot take one can.

So what should you actually do

The useful question is not "are statins good or bad." It is **"what is my actual risk, and what does the evidence say at that level of risk."**

1. **Get your risk calculated** with a validated calculator, using your real numbers, including ApoB and Lp(a) from the lab guide. Free, and it is the single most decision-relevant thing in this category.

2. **Ask which prevention you are in.** Secondary prevention and true primary prevention are different conversations with different evidence, and you are entitled to be told which one applies to you.
3. **Ask for absolute numbers, not relative ones.** "Reduces risk by 25%" is a relative figure. Ask what it means for someone with your risk over ten years, in whole people. Guidelines increasingly recognize that risk-based selection alone does not capture who actually benefits, and that individualized benefit estimates are better suited to shared decision-making (PMID 40694580).
4. **If muscle symptoms are your barrier**, ask about a structured rechallenge, a different statin, or a lower dose, because the evidence above says that conversation has good odds.
5. **Do everything in the diet guide regardless.** It is the part that helps at every risk level and it is not in competition with anything.

What this guide will not do is tell you to stop a prescription. If you are in secondary prevention, the evidence is strong and stopping is genuinely dangerous. If you are in primary prevention and you and your physician decide the absolute benefit does not justify it for you, that is a legitimate, evidence-consistent decision to make **together**, and it is a completely different act from quitting on the strength of a wellness article.

What outranks everything you can buy

The diet guide covers these in full and they are worth naming here:

- **The eating pattern**, where a network meta-analysis found comprehensive lifestyle modification, aerobic exercise, isometric training, low-sodium and high-potassium salt, salt restriction, breathing-control, meditation, and low-calorie diet all showing clear effects on blood pressure reduction (PMID 32975166).
- **Weight, if you carry extra**, particularly for fatty liver, where sustained weight loss is associated with histological improvement (PMID 41804193).
- **Treating sleep apnea**, which drives blood pressure and is commonly missed.

How to read this guide

Test first. ApoB and Lp(a) change what matters here, and the lab guide covers them.

Give lipid changes eight to twelve weeks before retesting, and blood pressure four weeks.

One change at a time.

Tier 1: red yeast rice, and the thing nobody tells you about it

This is the most requested supplement in this category and it needs the most honest handling.

What the evidence shows on efficacy and safety. A systematic review and meta-analysis of randomized trials found red yeast rice produced a reduction of about **1.02 mmol/L in LDL cholesterol**, which is roughly 39 mg/dL and is a genuinely substantial effect, while concluding that **safety is uncertain** and that it might be a safe and effective option for statin-intolerant patients only once the mild adverse reaction profile is confirmed in studies with adequate safety methodology (PMID 25897793). A later systematic review and meta-analysis focused specifically on safety concluded that red yeast rice used as a lipid-lowering dietary supplement seems overall tolerable and safe in moderately hypercholesterolaemic subjects (PMID 30844537).

Now the part that is usually left out, and it is the most important sentence in this guide.

The active compound in red yeast rice is **monacolin K, which is chemically identical to lovastatin**, a prescription statin.

So if you are taking red yeast rice specifically in order to avoid taking a statin, **you are taking a statin.** You are taking it at a dose that varies between products and batches, without prescription oversight, without the liver and muscle monitoring that comes with the prescription, and in a product category where content is not verified before sale.

That is not an argument against red yeast rice. It is an argument for knowing what you are doing. If red yeast rice is the route you and your clinician choose, the same cautions apply as to any statin: it interacts with the same drugs, it warrants the same monitoring, and it should not be combined with a prescription statin.

Cautions. Do not combine with a prescription statin. Do not use in pregnancy or nursing, or with liver disease. It interacts with the same drugs statins do, including some antibiotics, antifungals, and grapefruit. Tell every prescriber you take it, because it will not be on your medication list otherwise. Some products have historically been found contaminated with citrinin, a nephrotoxin, which is a specific reason to insist on third-party testing here.

Tier 2: omega-3, where the form decides the answer

This is the entry where most guides get it wrong by lumping all fish oil together.

What the evidence shows. A review of the major cardiovascular omega-3 trials concludes that **icosapent ethyl, a purified high-dose EPA form, improves cardiovascular outcomes among patients with elevated triglycerides at high cardiovascular risk, but EPA and DHA together does not.** Current guidelines endorse icosapent ethyl in statin-treated patients at high cardiovascular risk with triglycerides above 135 mg/dL (PMID 36562280).

So the honest translation. The cardiovascular outcome benefit belongs to a specific prescription product, in a specific population, on top of a statin. **It does not transfer to the combined EPA and DHA fish oil capsule on the shelf.** If you have high triglycerides and high cardiovascular risk, the conversation to have is about icosapent ethyl with your physician, not about buying more fish oil.

What over-the-counter fish oil is still reasonable for. Lowering triglycerides, and general dietary omega-3 intake. Not for the outcome claim above.

Cautions. Higher doses increase bleeding tendency, which matters with anticoagulants, antiplatelet drugs, and before surgery.

Tier 3: the supporting cast, honestly graded

Soluble fiber. Psyllium and beta-glucan lower LDL modestly and reliably, and they are cheap, safe, and do other useful things. Start low to avoid bloating and take with plenty of water. This is the least glamorous and most defensible item in the guide.

Plant sterols and stanols. Modest additional LDL lowering. Reasonable as an addition.

Magnesium and potassium are covered in the diet guide, because food is the better route and potassium supplements carry real risk in kidney disease and with certain blood pressure medications. **Do not supplement potassium without your clinician**, particularly on an ACE inhibitor, ARB, or potassium-sparing diuretic.

CoQ10 for statin muscle symptoms, where two meta-analyses disagree

This deserves its own section because you will be told it works, and the evidence is genuinely split.

The negative one. A systematic review and meta-analysis assessing whether CoQ10 improves statin-associated myalgia and helps people stay on statins **did not demonstrate benefit** for muscle pain, and found no improvement in the proportion of patients remaining on statin treatment, with a risk ratio of 0.99 (PMID 32179207).

The positive one. A 2025 systematic review and meta-analysis of CoQ10 for myopathy in statin-treated patients found a reduction in muscle pain, with a weighted mean difference of -0.96 and a confidence interval of -1.88 to -0.03, which is statistically significant but with the upper bound sitting very close to zero. The authors state more research is needed for evidence-based recommendations (PMID 41158831).

How I read that. Two meta-analyses on the same question reaching different conclusions, with the positive one's confidence interval nearly touching no-effect. CoQ10 is safe and inexpensive, and it is a reasonable thing to try if muscle symptoms are your barrier to staying on a statin, which is a worthwhile goal. Just do not expect much, and do not let trying it delay the conversation about rechallenge or a different statin, which has better evidence behind it.

Cautions. CoQ10 can reduce the effect of warfarin. It may modestly lower blood pressure, relevant if you are treated for that.

What I deliberately left out, and why

- **Anything sold as a statin replacement.** Covered above. The blinded rechallenge data say most people who think they cannot tolerate a statin can.
- **Combined EPA and DHA fish oil for cardiovascular outcome prevention.** The outcome benefit belongs to purified high-dose EPA as a prescription, and did not transfer to the combination (PMID 36562280).
- **Niacin at cardiovascular doses.** Outcome trials were disappointing and the side effect burden is real. This is a prescriber conversation, not a shelf purchase.
- **Chelation therapy for atherosclerosis.** Not supported for this use, and not benign.
- **Nattokinase and serrapeptase for "cleaning arteries."** No outcome evidence, and real bleeding-risk questions.
- **High-dose antioxidant vitamins for cardiovascular prevention.** Repeatedly tested, repeatedly disappointing.
- **Anything that encourages you to skip a risk calculation.** Your risk number, produced with a validated calculator and your actual values, is the single most decision-relevant thing in this category, and it is free.

Telling you what to skip is part of what you paid for.

Running a fair trial

Weeks 1 to 4, baseline. Home blood pressure twice daily for a week, waist measurement, and the labs from the lab guide.

Weeks 5 to 16, one change at a time, starting with the diet guide rather than a bottle.

Week 17, retest. Lipids need eight to twelve weeks to reflect a change. Blood pressure responds faster.

Realistic expectations

What is realistic. Modest LDL reduction from soluble fiber and sterols. Meaningful blood pressure change from the diet and movement work. Substantial LDL reduction from red yeast rice, because it contains a statin.

What is not realistic. A supplement replacing appropriate medical treatment in someone at genuine risk. Reversing established plaque with a capsule.

Cost sense. The highest-value items here are a home blood pressure monitor, a tape measure, and psyllium. Together they cost less than one month of most supplement stacks.

When to seek professional care

Emergency: chest pain or pressure, one-sided weakness or speech difficulty, sudden severe breathlessness, tearing chest or back pain. Call 911.

Soon: home blood pressure repeatedly at or above 140/90, chest discomfort on exertion, new breathlessness on stairs, swelling in both ankles, or muscle pain that is severe or comes with dark urine, which needs same-day assessment.

Where to find quality versions of these

Nothing here names a brand, on purpose.

Third-party testing matters more here than almost anywhere. Red yeast rice is the specific case: monacolin content varies between products, and citrinin contamination has been documented. If a red yeast rice product cannot show you independent testing, do not take it.

Match the form to the evidence. Purified high-dose EPA is a prescription, not a shelf product. Psyllium and beta-glucan are the studied soluble fibers.

If you would rather not evaluate all of that yourself, I keep a screened dispensary through Fullscript, which ships directly to you. I hold no physical inventory. It is screened for the standards above.

> ***Root Health dispensary: us.fullscript.com/welcome/drseay***
>
> *Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the three highest-value items in this guide, a blood pressure cuff, a tape measure, and taking your prescribed medication, earn us nothing.*

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your numbers, your calculated risk, and your medication list.

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> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=heart-supplement***

If you would rather start on your own, start with home blood pressure and a waist measurement:

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Read the companion documents alongside this one. **Understanding Your Heart and Metabolic Labs** covers ApoB and Lp(a), and **The Heart and Metabolic Diet Guide** covers the eating pattern that outperforms most of this page.

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